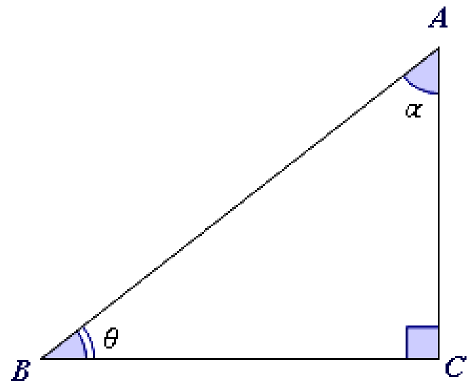
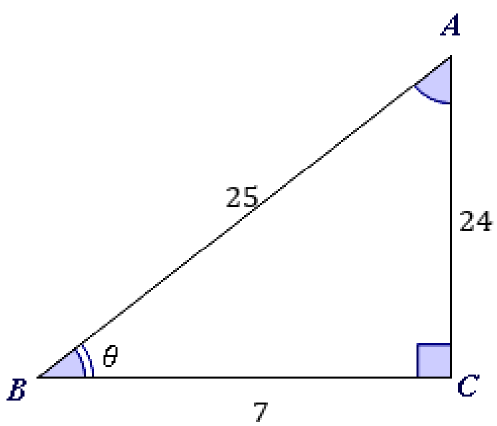
Sine, cosine, and tangent ratios

1 Consider the following right triangle. Identify the angle that is opposite the hypotenuse.

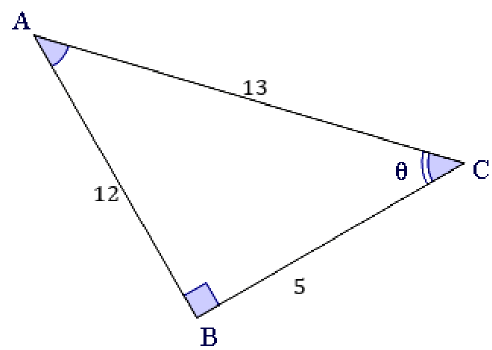


2 Using the reference angle θ, find the value of the ratio $\frac{\text{opposite}}{\text{adjacent}}$ of the following triangles.

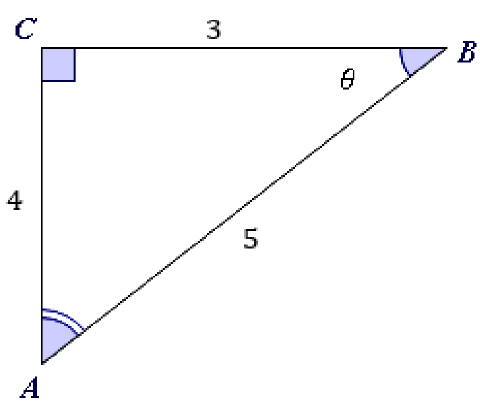
a



b



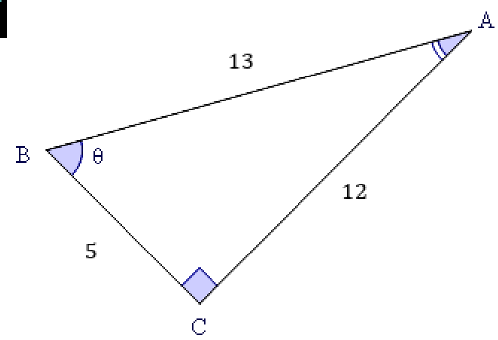
3 Using the reference angle θ, find the value of the ratio $\frac{\text{opposite}}{\text{hypotenuse}}$ for the given triangle.



4 Consider the following right triangle. Using the reference angle θ, find the following ratios:

a $\dfrac{\text{adjacent}}{\text{hypotenuse}}$

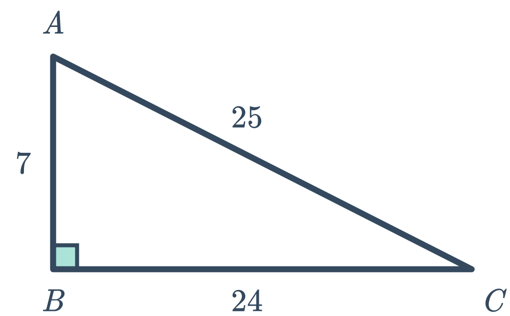
b $\dfrac{\text{opposite}}{\text{hypotenuse}}$



5 Consider the given triangle.

a For $\angle ACB$, find the value of the ratio $\dfrac{\text{opposite}}{\text{hypotenuse}}$.

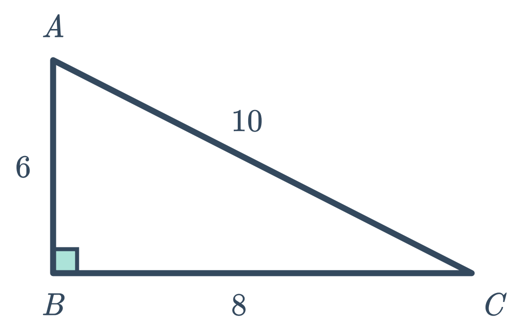
b For $\angle CAB$, find the value of the ratio $\dfrac{\text{adjacent}}{\text{hypotenuse}}$.



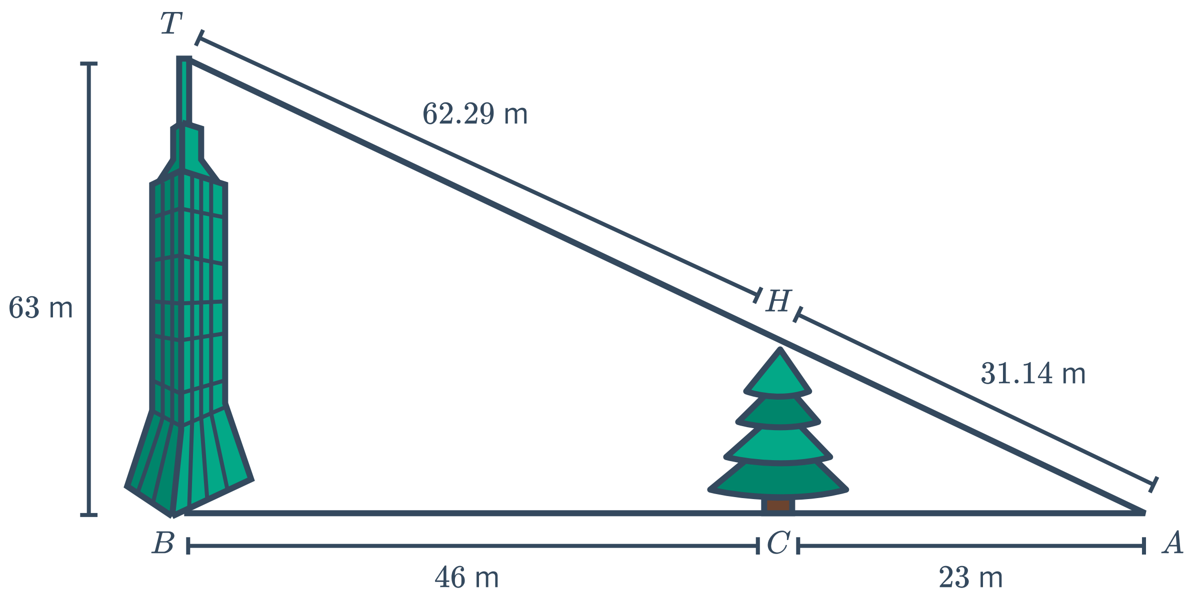
6 Consider the given triangle.

a For $\angle ACB$, find the value of the ratio $\dfrac{\text{adjacent}}{\text{hypotenuse}}$.

b For $\angle CAB$, find the value of the ratio $\dfrac{\text{opposite}}{\text{hypotenuse}}$.



7 Consider the figure below.



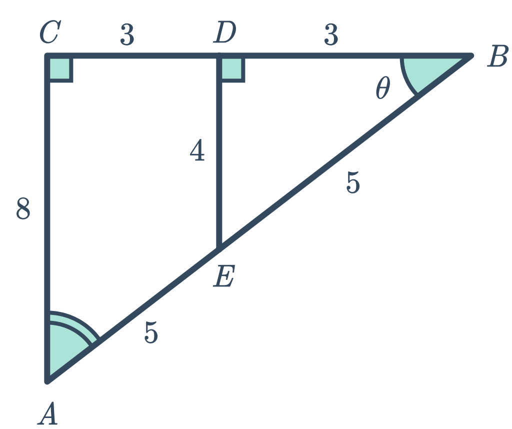
a Using the triangle created by the building, find the ratio of the opposite side to the adjacent side of $\angle TAB$.

b The tree has a height of 21 meters. For the smaller triangle created by the tree, find the ratio of the opposite side to the adjacent side of $\angle HAC$.

c Using the triangle created by the tree, determine the ratio of the adjacent side to the hypotenuse for $\angle HAC$.

d Using the triangle created by the building, determine the ratio of the opposite side to the hypotenuse for $\angle TAB$.

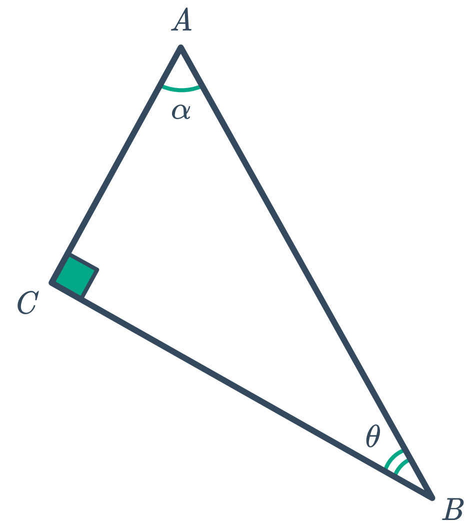
8 Consider the following diagram.
Using the reference angle θ, state the ratio of sides that is equivalent to $\frac{4}{5}$.



9 Write down the following ratios for the given triangle:

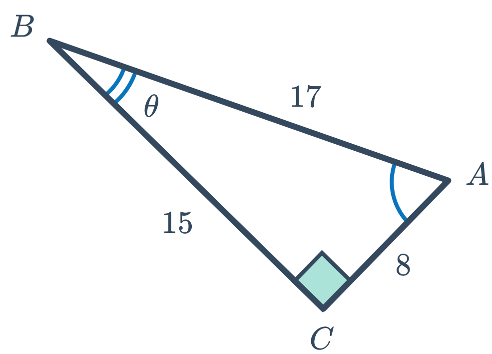
a $\sin\theta$ **b** $\cos\theta$

c $\tan\theta$



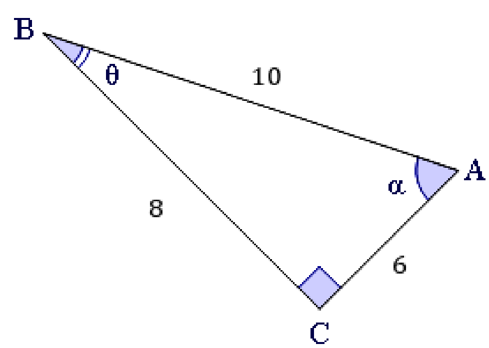
10 Using the information in the given triangle, find the values of the following ratios:

a $\tan\theta$ **b** $\cos\theta$



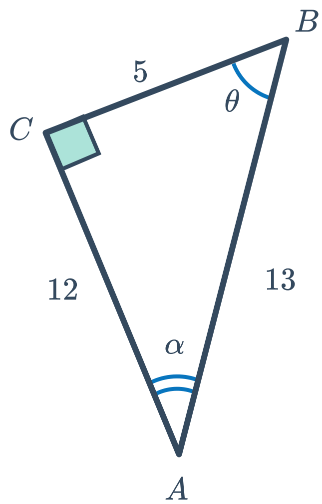
11 Using the information in the given triangle, find the values of the following ratios:

a $\tan\theta$ **b** $\sin\theta$

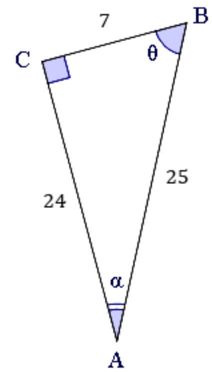


12 For each of the following triangles, find the [redacted] of the given ratio:

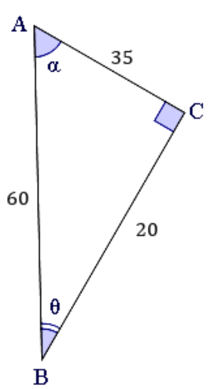
a $\cos\theta$



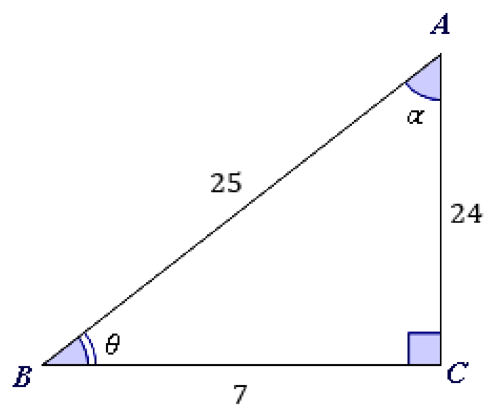
b $\sin\theta$



c $\tan\theta$



d $\tan\theta$

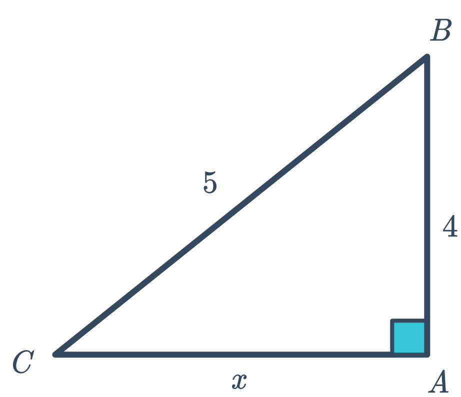


15 In the following triangle, $\sin\theta = \dfrac{4}{5}$.

a Identify which angle is represented by θ.

b Find the value of $\cos\theta$.

c Find the value of $\tan\theta$.



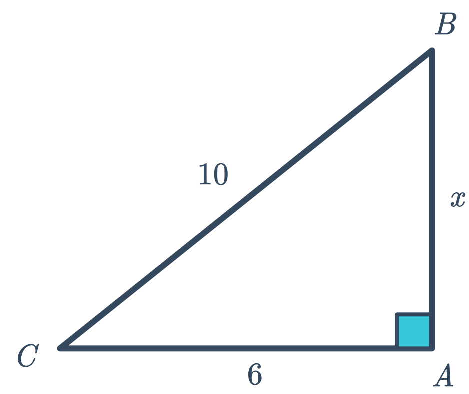
16 For the given triangle, $\cos\theta = \dfrac{6}{10}$.

a Identify the angle represented by θ.

b Find the value of x.

c Find the value of $\sin\theta$.

d Find the value of $\tan\theta$.

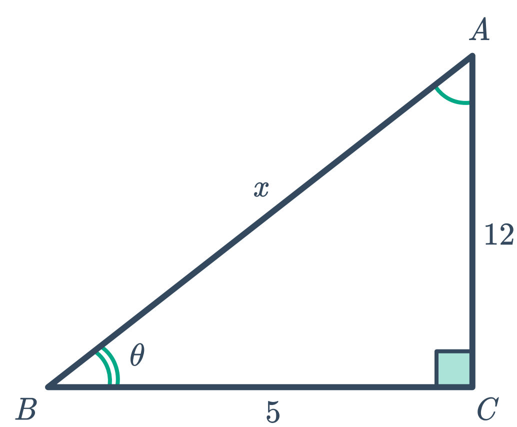


17 Consider the following triangle.

a Find the value of x.

b Find the value of $\sin\theta$.

c Find the value of $\cos\theta$.

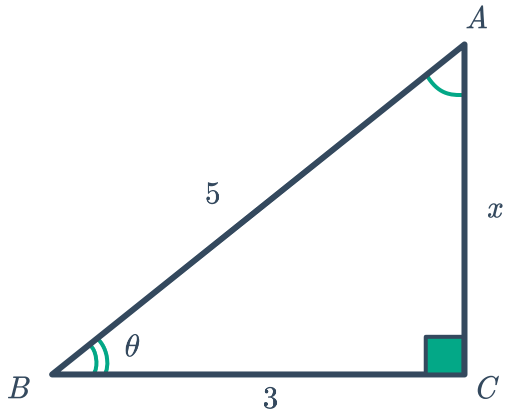


18 For each of the following triangles:

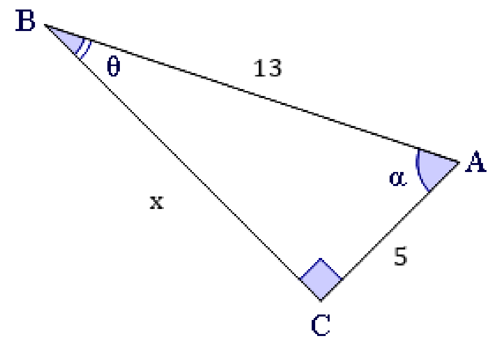
i Find the value of x.

ii Hence, find the value of $\tan\theta$.

a



b



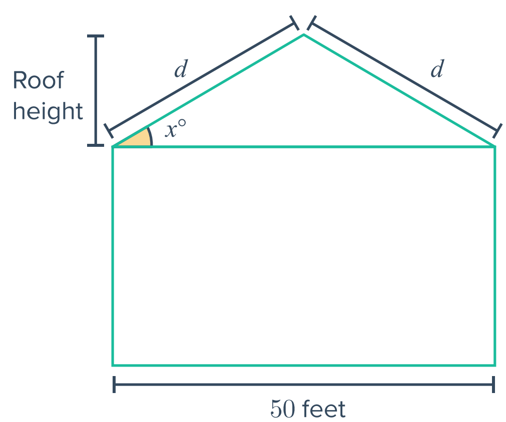
19 Determine whether the following pairs of right triangles are similar.

a A pair of corresponding angles are found to have a sine ratio of $\dfrac{240}{250}$ and $\dfrac{24}{25}$.

b A pair of corresponding angles are found to have a tangent ratio of $\dfrac{140}{160}$ and $\dfrac{70}{80}$.

c A pair of corresponding angles are found to have a tangent ratio of $\dfrac{320}{420}$ and $\dfrac{316}{416}$.

20 Beth is building a house, as shown.
She wants the roof height to be between 3.5 ft and 5 ft. She must decide the angle measure to use for the slant of the roof when the slant height is d ft.
State the inequality that Beth can use in order to ensure that her roof height will be within the desired range.



Additional questions

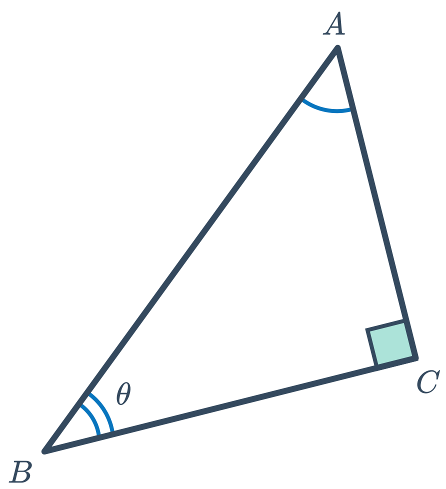
21 For the following triangles, with reference angle θ, identify:

i The adjacent side

ii The opposite side

iii The hypotenuse

a



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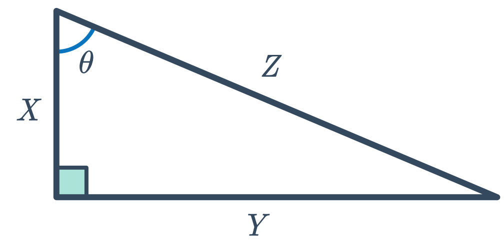
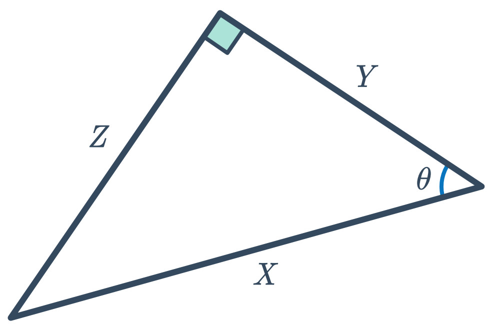
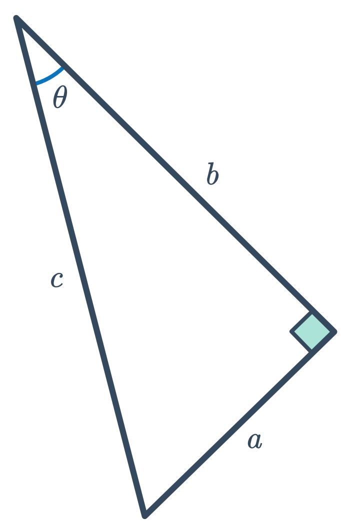


c

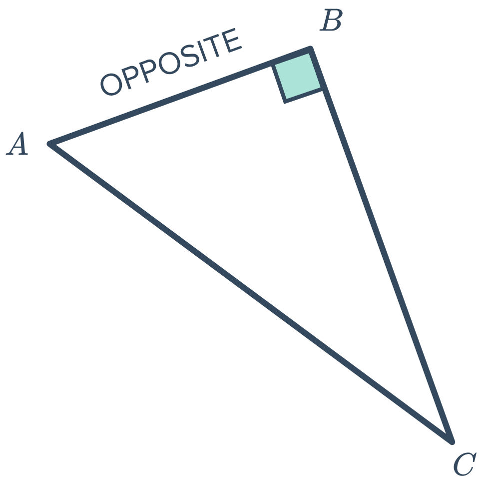


d



22 Identify the reference angle in each of the f[redacted]ing triangles given the labelled side:

a



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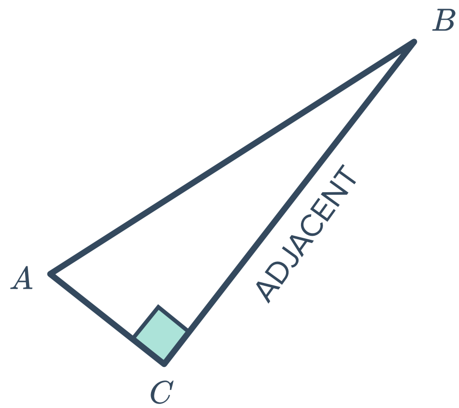

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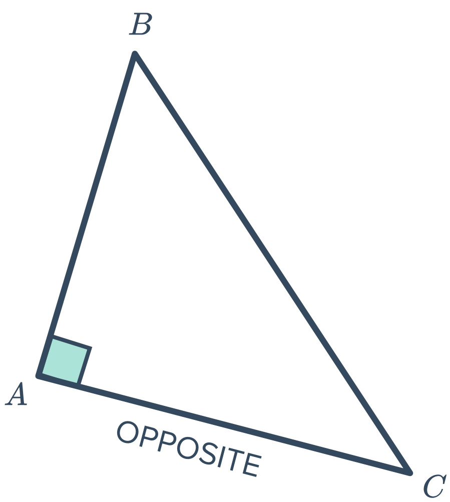

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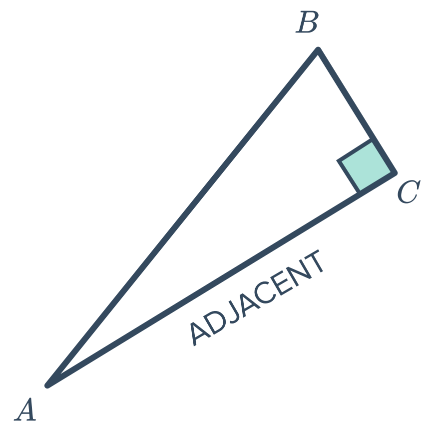


23 Consider the following triangle:

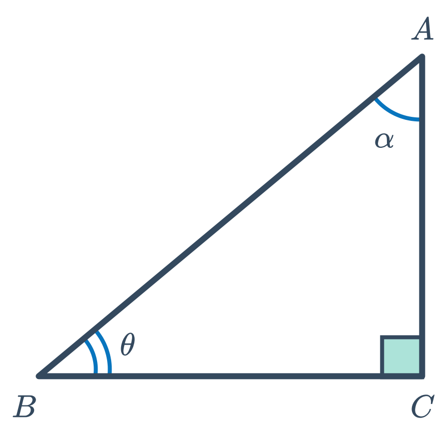
a State the opposite side to angle θ.

b State the adjacent side to angle θ.

c State the opposite side to angle α.

d State the adjacent side to angle α.

e Identify the angle that is opposite the hypotenuse.



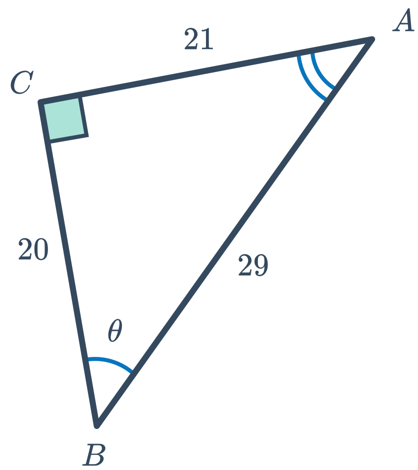
24 For each of the following triangles:

i Find the value of $\sin\theta$.

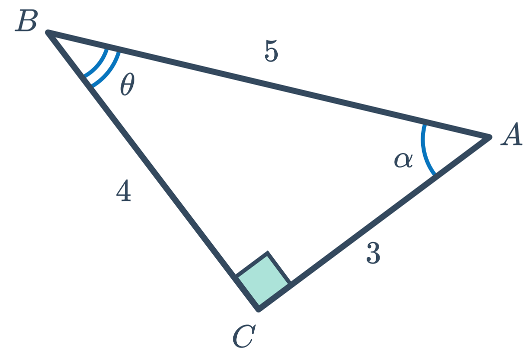
ii Find the value of $\cos\theta$.

iii Find the value of $\tan\theta$.

a



b



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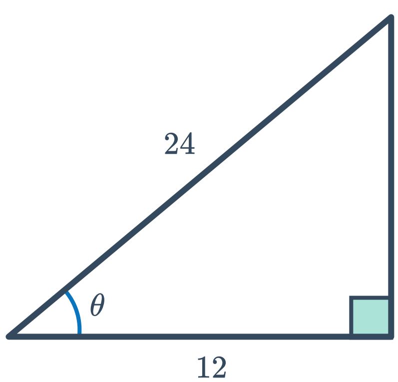
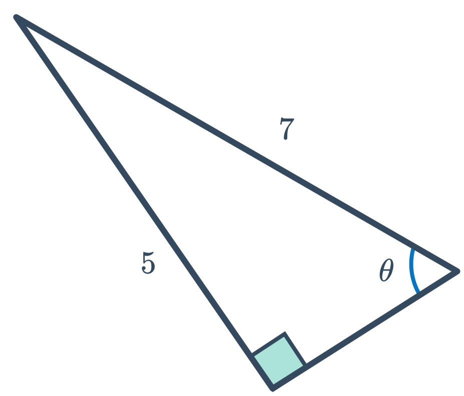
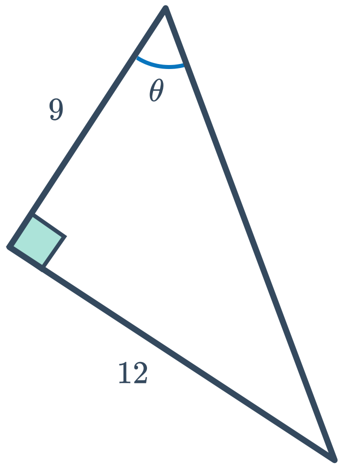
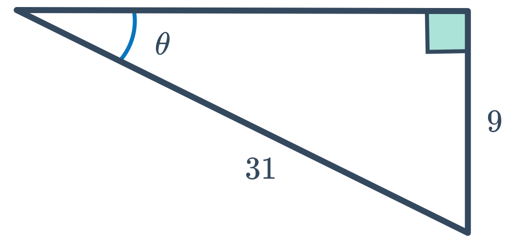


d



e



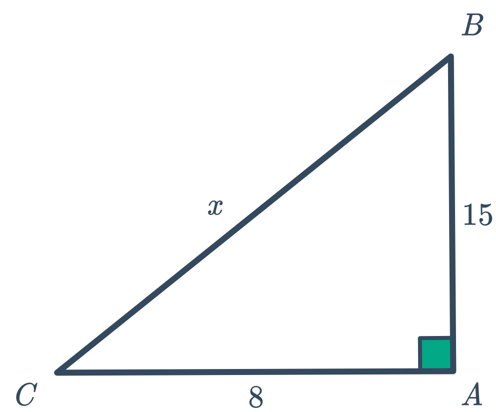


25 In the following triangle $\tan\theta = \dfrac{15}{8}$.

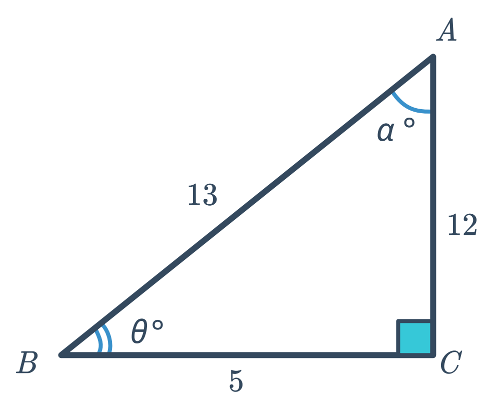
a Which angle is represented by θ?

b Find $\cos\theta$.

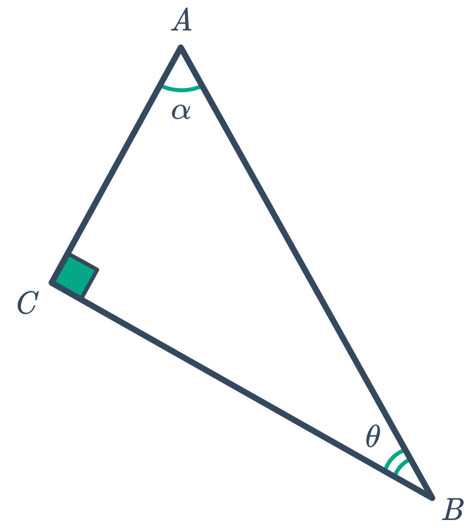
c Find $\sin\theta$.



26 With reference to the diagram, show that $\dfrac{\sin\theta}{\cos\theta} = \tan\theta$.



27 Write down the following ratios for the given triangle:

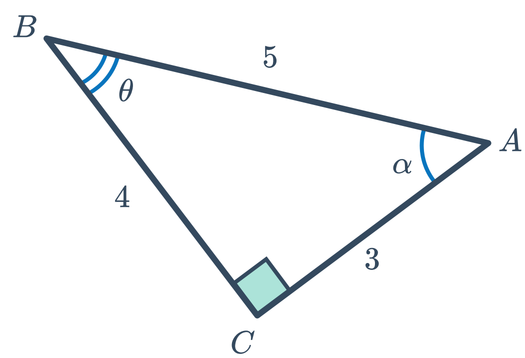


a $\sin \alpha$

b $\cos \alpha$

c $\tan \alpha$

28 With reference to the diagram, does $\sin \theta = \cos \alpha$? Explain why or why not.



29 A right triangle is enlarged by a factor of 4. Explain why the trigonmetric ratios are the same for both triangles.

30 Explain why $\tan 30^\circ = \dfrac{\sqrt{3}}{3}$. Include a diagram with your explanation.